

Software Requirements Specification (SRS)

Project: Electronics Store Inventory Management System

Version: 1.0

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Revision History

Version	Date	Author	Change Summary	Approval
1.0	06-09-2026	Vaishnavi Bandaru, Valmiki Uma, Yerramilli Siva Sai Saharsh, Yashas	Initial SRS	

Approvals

Role	Name	Signature / Email	Date
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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for an Electronics Store Inventory Management System. The system is intended to help electronics stores efficiently manage their products, inventory levels, suppliers, purchases, sales, and stock movements through a centralized software application.

The SRS defines the functional requirements, non-functional requirements, interfaces, security requirements, and acceptance criteria of the proposed system. It serves as a reference for the development, testing, and evaluation of the project.

1.2 Scope

The Electronics Store Inventory Management System is designed to assist store personnel in managing electronic products and their inventory.

The system will allow authorized personnel to:

- Manage product information.
- Track available stock.
- Manage supplier information.
- Record purchases and update inventory accordingly.
- Record sales and update inventory accordingly.
- Search for products.
- Identify low-stock products.
- Generate basic inventory and sales reports.

The system is intended for internal store use and does not provide a customer-facing shopping or e-commerce interface. Advanced accounting, online payment processing, delivery management, and external e-commerce integration are outside the scope of the project.

1.3 Audience

The intended audience for this document includes:

- Project developers

- Project team members
- Store administrators
- Store employees
- System testers
- Project evaluators/instructors
- Future system maintainers

1.4 Definitions and Acronyms

Term	Definition
SRS	Software Requirements Specification
UI	User Interface
DB	Database
SKU	Stock Keeping Unit
Admin	System administrator
Inventory	Collection of products and their available stock
Supplier	Entity that supplies electronic products to the store
Reorder Level	Minimum stock quantity at which replenishment is required
CRUD	Create, Read, Update, Delete
API	Application Programming Interface

2. Overall Description

2.1 Product Perspective

The Electronics Store Inventory Management System will be a centralized application connected to a database that stores information about products, categories, suppliers, users, purchases, sales, and inventory transactions.

The system will provide different interfaces depending on the user's role. Administrators will have access to management and reporting functions, while store employees will have access to operational functions such as viewing products, recording sales, and updating stock.

The system will automatically maintain inventory information based on recorded purchase and sales transactions.

2.2 Major Product Functions

The system shall provide the following major functions:

- User authentication
- Product management
- Inventory tracking
- Supplier management
- Purchase recording
- Sales recording
- Automatic stock updates
- Low-stock identification
- Product search
- Basic inventory and sales reporting

2.3 User Roles and Characteristics

Store Manager / Administrator:

The primary user responsible for managing products, suppliers, inventory, purchases, sales, and reports. This user has access to all system functions.

Store Staff/ Sales Staff:

Users responsible for day-to-day operations such as viewing products, checking stock, and recording sales. Access to administrative functions may be restricted.

2.4 Operating Environment

The system is expected to operate in the following environment:

- Desktop or laptop computer
- Modern web browser or supported desktop interface
- Application server/runtime environment
- Relational database management system
- Local or network-based environment

- Internet/network connection where required

The exact programming language, framework, and database technology will be determined during the system design phase.

2.5 Constraints

The system is subject to the following constraints:

- Only authorized users shall be permitted to access protected functions.
- Inventory quantities shall not become negative.
- Each product shall have a unique product identifier/SKU.
- The database shall maintain consistency between product, inventory, purchase, and sales records.
- The system shall use a relational database for structured inventory and transaction data.
- Access to administrative functions shall be restricted according to user roles.

3. External Interface Requirements

3.1 User Interfaces

The system shall provide a simple interface for authorized store personnel. The main pages shall include:

- **Login Page** – Authentication of authorized staff.
- **Dashboard** – Overview of products, stock levels, low-stock items, and recent transactions.
- **Products & Inventory Page** – View, add, update, and manage product and stock information.
- **Suppliers & Purchases Page** – Manage supplier information and record purchases.
- **Sales Page** – Record sales and automatically update inventory.
- **Reports Page** – View basic inventory, sales, and purchase reports.

3.2 Hardware Interfaces

The system does not require specialized hardware.

It shall operate using standard:

- Desktop computers
- Laptop computers
- Keyboard
- Mouse/touchpad
- Display monitor

Optional hardware such as barcode scanners may be considered as a future enhancement but is outside the core scope of this project.

3.3 Software Interfaces

The system shall interact with:

- Relational database management system for persistent storage.
- Application/backend services for business logic.
- Frontend/user interface for user interaction.

If required, the system may expose internal REST APIs between the frontend and backend.

3.4 Communication Interfaces

If implemented as a web-based application:

- Communication between client and server shall use HTTP/HTTPS.
- Database communication shall occur through the backend application.
- Sensitive information shall not be transmitted unnecessarily.

4. System Features

Each requirement below includes an acceptance criterion and test case reference. Requirement IDs follow the format ESIMS-F-###.

4.1 Authentication

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Comments
ESIMS-F-001	The system shall allow authorized	Functional	High	Store Staff	Valid credentials result in	—

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Comments
	store personnel to log in using valid credentials.				successful login. Test: TC-Auth-01	
ESIMS-F-002	The system shall reject login attempts containing invalid credentials.	Functional	High	Security	Invalid credentials result in an error message. Test: TC-Auth-02	—

4.2 Product & Inventory Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Comments
ESIMS-F-003	The system shall allow authorized personnel to add a new electronic product with details such as name, category, price, and quantity.	Functional	High	Store Staff	A valid product is successfully added. Test: TC-Prod-01	—
ESIMS-F-004	The system shall allow authorized personnel to	Functional	High	Store Staff	Updated product information is saved	—

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Comments
	update existing product information.				correctly. Test: TC-Prod-02	
ESIMS-F-005	The system shall allow personnel to view the current stock quantity of each product.	Functional	High	Store Staff	Current stock quantity is displayed correctly. Test: TC-Inv-01	—
ESIMS-F-006	The system shall allow personnel to search for products using product name or product ID/SKU.	Functional	Medium	Store Staff	Matching products are displayed. Test: TC-Inv-02	—
ESIMS-F-007	The system shall identify products whose stock is at or below the configured reorder level.	Functional	High	Store Staff	Low-stock products are displayed. Test: TC-Inv-03	—
ESIMS-F-008	The system shall prevent inventory quantity from	Functional	High	Business Rule	A transaction exceeding available	—

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Comments
	becoming negative.				stock is rejected. Test: TC-Inv-04	

4.3 Supplier & Purchase Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Comments
ESIMS-F-009	The system shall allow personnel to add and update supplier information.	Functional	Medium	Store Staff	Supplier information is stored or updated successfully. Test: TC-Sup-01	—
ESIMS-F-010	The system shall allow personnel to record a purchase from a supplier.	Functional	High	Store Staff	Purchase transaction is successfully recorded. Test: TC-Pur-01	—
ESIMS-F-011	The system shall automatically increase the corresponding product stock when a purchase is recorded.	Functional	High	Inventory	Stock increases by the purchased quantity. Test: TC-Pur-02	—

4.4 Sales Management

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Comments
ESIMS-F-012	The system shall allow personnel to record a sale containing one or more products.	Functional	High	Store Staff	A valid sale is successfully recorded. Test: TC-Sale-01	—
ESIMS-F-013	The system shall calculate the total amount of a sale based on product prices and quantities.	Functional	High	Business Rule	Calculated total matches the expected amount. Test: TC-Sale-02	—
ESIMS-F-014	The system shall automatically decrease product stock after a successful sale.	Functional	High	Inventory	Stock decreases by the quantity sold. Test: TC-Sale-03	—

4.5 Reports

Req ID	Requirement	Type	Priority	Source/Stakeholder	Acceptance Criteria / Test Case	Comments
ESIMS-F-015	The system shall generate a basic inventory report showing products and their current stock levels.	Functional	Medium	Store Staff	Inventory report displays current stock information. Test: TC-Report-01	—
ESIMS-F-016	The system shall generate a sales report for a selected date or date range.	Functional	Medium	Store Staff	Report displays sales within the selected period. Test: TC-Report-02	—

5. Non-Functional Requirements

The following requirements define the quality attributes and operational characteristics of the system. IDs follow ESIMS-NF-###.

Req ID	Requirement	Category	Priority	Acceptance Criteria
ESIMS-NF-001	The system shall respond to common operations such as product search and inventory lookup within 3 seconds under normal load.	Performance	High	90% of tested requests complete within 3 seconds.
ESIMS-NF-002	The system shall maintain consistent inventory quantities	Reliability	High	Inventory values remain correct after tested transactions.

Req ID	Requirement	Category	Priority	Acceptance Criteria
	after purchase and sales transactions.			
ESIMS-NF-003	The system shall be available during the store's intended operating hours except during planned maintenance.	Availability	Medium	System remains accessible during testing.
ESIMS-NF-004	The system shall provide a simple and consistent user interface that allows store personnel to perform common operations easily.	Usability	Medium	Users can complete common tasks without critical usability errors.
ESIMS-NF-005	The system shall protect stored user and transaction information from unauthorized access or modification.	Security	High	Unauthorized access and modification attempts are rejected.

5.1 Security

5.1.1 Security Objectives

The system shall satisfy the following security objectives:

Security Objective 1 – Confidentiality:

The system shall ensure that user credentials and store transaction information are accessible only to authorized personnel.

Security Objective 2 – Integrity:

The system shall prevent unauthorized modification of product, inventory, purchase, and sales information.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance Criteria / Test Case
ESIMS-SR-001	The system shall require users to authenticate before accessing protected functions.	Security	High	Unauthenticated users cannot access the system. Test: TC-Sec-01
ESIMS-SR-002	The system shall restrict administrative functions to authorized personnel.	Security	High	Unauthorized users cannot access restricted functions. Test: TC-Sec-02
ESIMS-SR-003	The system shall store passwords using secure hashing rather than plaintext.	Security	High	Passwords are not stored as plaintext. Test: TC-Sec-03
ESIMS-SR-004	The system shall validate user input before processing or storing it.	Security	High	Invalid input is rejected appropriately. Test: TC-Sec-04
ESIMS-SR-005	The system shall prevent unauthorized modification or deletion of inventory and transaction records.	Security	High	Unauthorized modification/deletion attempts are rejected. Test: TC-Sec-05

6. Quality Attributes & Acceptance Tests

The system shall be evaluated based on the following quality attributes:

6.1 Performance

The system shall provide timely responses for common operations such as product search, inventory lookup, and recording sales and purchases.

6.2 Reliability

The system shall maintain consistent inventory quantities after purchase and sales transactions. Failed transactions shall not result in incorrect or partial inventory updates.

6.3 Usability

The system shall provide clear menus, forms, tables, and feedback messages to allow store personnel to perform common operations easily.

6.4 Security

Only authenticated and authorized store personnel shall be able to access protected system functionality.

6.5 Maintainability

The system shall use a modular structure so that components such as product management, inventory management, sales, purchases, and reporting can be modified with minimal impact on other components.

Acceptance Criteria

The project shall be considered acceptable when:

- All high-priority functional requirements have been implemented and verified.
- All high-priority requirements have corresponding test cases.
- Inventory quantities remain accurate after purchase and sales transactions.
- Authentication and authorization tests pass for both defined user roles.
- All defined security requirements have been verified.
- Major store workflows operate successfully.
- The Requirements Traceability Matrix is complete.
- No critical defects remain at the time of submission.

Acceptance Test Suites

The following test suites shall be performed:

- Authentication and authorization testing
- Product management testing
- Supplier management testing
- Inventory testing
- Purchase transaction testing
- Sales transaction testing
- Search and filtering testing

- Report generation testing
- Performance testing
- Security testing
- Usability testing

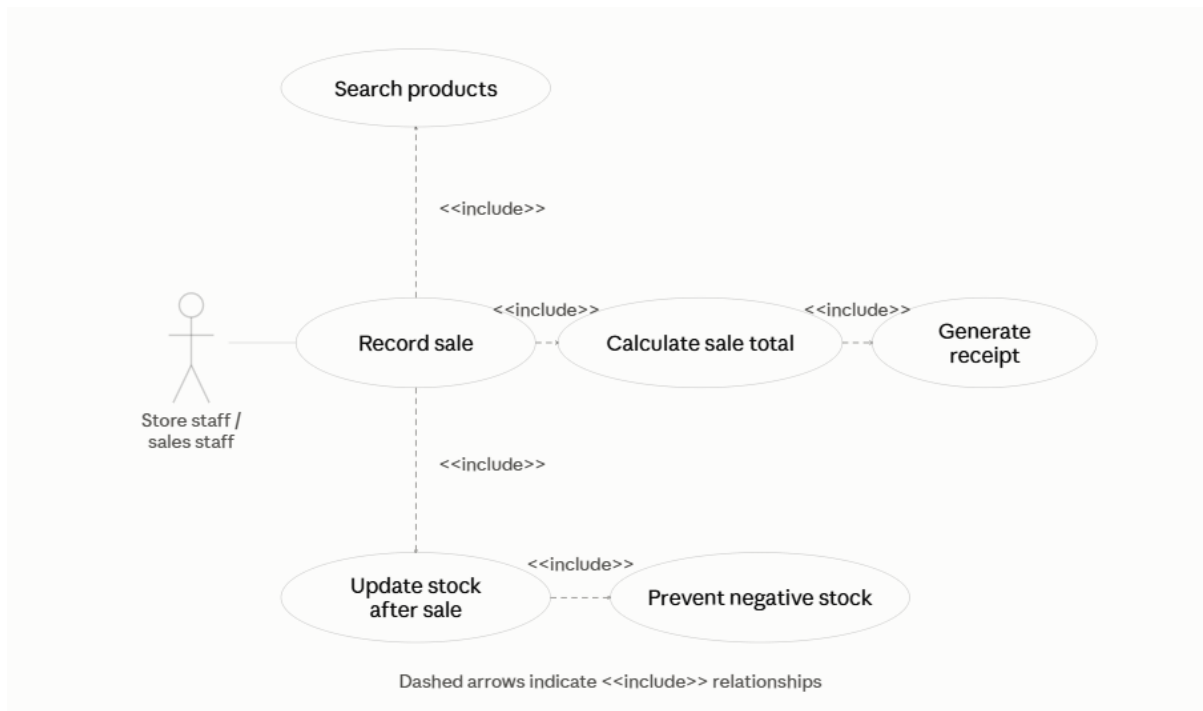
7. System Models and Diagrams

7.1 UML Use-Case Diagram

Overall System Use-Case Diagram:



Sales & Inventory Use-Case Diagram:



8. Requirements Traceability Matrix (RTM)

The Requirements Traceability Matrix (RTM) maps each system requirement to the corresponding system module and test case. It will be updated throughout the development and testing phases.

Req ID	Requirement Short	Section Ref.	Module	Test Case(s)	Status	Comments
ESIMS-F-001	User login	4.1	Authentication Module	TC-Auth-01	N	—
ESIMS-F-002	Invalid login handling	4.1	Authentication Module	TC-Auth-02	N	—
ESIMS-F-003	Add product	4.2	Product Module	TC-Prod-01	N	—
ESIMS-F-004	Update product	4.2	Product Module	TC-Prod-02	N	—
ESIMS-F-005	View stock	4.2	Inventory Module	TC-Inv-01	N	—

Req ID	Requirement Short	Section Ref.	Module	Test Case(s)	Status	Comments
ESIMS-F-006	Search products	4.2	Product Module	TC-Inv-02	N	—
ESIMS-F-007	Low-stock detection	4.2	Inventory Module	TC-Inv-03	N	—
ESIMS-F-008	Prevent negative stock	4.2	Inventory Module	TC-Inv-04	N	Business rule
ESIMS-F-009	Manage suppliers	4.3	Supplier Module	TC-Sup-01	N	—
ESIMS-F-010	Record purchase	4.3	Purchase Module	TC-Pur-01	N	—
ESIMS-F-011	Update stock after purchase	4.3	Inventory Module	TC-Pur-02	N	—
ESIMS-F-012	Record sale	4.4	Sales Module	TC-Sale-01	N	—
ESIMS-F-013	Calculate sale total	4.4	Sales Module	TC-Sale-02	N	—
ESIMS-F-014	Update stock after sale	4.4	Inventory Module	TC-Sale-03	N	—
ESIMS-F-015	Inventory report	4.5	Reporting Module	TC-Report-01	N	—
ESIMS-F-016	Sales report	4.5	Reporting Module	TC-Report-02	N	—
ESIMS-NF-001	Response time	5	Application	TC-Perf-01	N	—
ESIMS-NF-002	Inventory consistency	5	Database	TC-Rel-01	N	—

Req ID	Requirement Short	Section Ref.	Module	Test Case(s)	Status	Comments
ESIMS-NF-003	Availability	5	Application	TC-Avail-01	N	—
ESIMS-NF-004	Usability	5	User Interface	TC-UX-01	N	—
ESIMS-NF-005	Data protection	5	Security Module	TC-Sec-01	N	—
ESIMS-SR-001	Authentication	5.1.2	Authentication Module	TC-Sec-02	N	—
ESIMS-SR-002	Access restriction	5.1.2	Authorization Module	TC-Sec-03	N	Two user roles
ESIMS-SR-003	Password hashing	5.1.2	Authentication Module	TC-Sec-04	N	—
ESIMS-SR-004	Input validation	5.1.2	Application	TC-Sec-05	N	—
ESIMS-SR-005	Transaction protection	5.1.2	Inventory/Sales Module	TC-Sec-06	N	—

Status

- **N** — Not yet implemented/tested
- **P** — Partially implemented/tested
- **A** — Accepted/verified